
How Do World Action Models Use Their Futures?

Evidence from Mechanistic Interventions

Nicole Ma

Abstract

World action models jointly generate possible future observations and robot actions, but joint generation does not show whether a particular future affects the action. We introduce reachable-donor future transplantation. From one restored simulator state, two unmodified policy runs produce alternative futures, actions, and endpoints. We replace the recipient’s future with the donor’s while holding the current observation, instruction, sampling noise, and denoising schedule fixed, then measure movement toward the donor action and its one-chunk simulator endpoint. In Cosmos 3, a frozen population study across 22 states and six RoboLab tasks finds predicted-future action steering of 0.992 and executed-future endpoint steering of 1.003 on a normalized recipient-to-donor scale. Replacing donor-run future-token keys and values with self-future counterparts reduces predicted-action steering by 0.877; action and endpoint reductions are positive in all 22 states. Yet isolated robot- and object-pixel interventions each have effects below 0.011. In Cosmos Policy’s direct-policy mode, coherent futures steer actions and endpoints at early states across all ten LIBERO-Long tasks, with visual content carrying most of the effect. The results establish donor-specific future use and an active internal pathway, while leaving candidate-future search and value-based outcome comparison untested.

1 Introduction

Many world action models (WAMs) generate future observations alongside robot actions [8, 14]. This architecture suggests a simple account: the model predicts what may happen and uses that prediction to act. Joint generation alone does not establish this computation. A generated future could be behaviorally irrelevant, serve as a general workspace, describe the robot motion already paired with the action, or represent task consequences used to choose among actions.

These alternatives imply different functional roles. A policy that conditions action inference on predicted future observations resembles an inverse-dynamics interface [1, 7]. A system that generates alternatives and compares their predicted outcomes with a cost or learned heuristic more closely resembles explicit visual planning [2, 3]. We study Cosmos Policy’s direct-policy mode and the diffusion-generator component of Cosmos3-Nano-Policy-DROID. The inference procedures evaluated here do not generate and score multiple candidate actions before acting; Cosmos Policy’s separate best-of- N planning mode is outside our study.

RIFT provides the closest causal evidence that future representations matter [22]. Masking the future read or editing values stored at future positions changes closed-loop trajectories and task success across several WAMs. This establishes dependence on a structured future interface, but a destructive intervention does not say what a particular coherent alternative future tells the policy to do. A damaged representation can move an action in any direction.

We make the test directional. From the same restored state, two native policy continuations define a recipient and a donor. We insert the donor future into the recipient computation while retaining the recipient’s current inputs and random draws. The donor action is never supplied to the model;

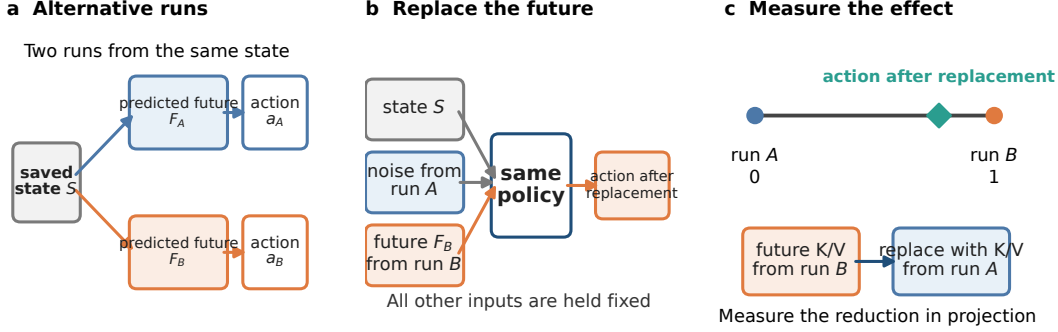


Figure 1: Reachable-donor future transplantation. Two unmodified continuations start from the same simulator state. The recipient is recomputed with the donor future while its other inputs and random draws remain fixed. Signed projections measure movement from recipient behavior (0) toward donor behavior (1); K/V replacement then tests whether future-token activations carry that shift.

it defines an independent direction against which we score the intervened action and its executed simulator endpoint (Figure 1). If the future only perturbs computation, the result need not align with the donor. If its content steers behavior, both action and endpoint should move toward the continuation paired with that future.

The experiments follow three linked questions. First, do coherent donor futures steer behavior? Second, which parts of the future recreate that effect? Third, which future-to-action interfaces carry it? Cosmos 3 provides the strongest within-model chain: steering generalizes across 22 sampled states, isolated robot or object pixels do not recreate it, and future-token K/V replacement suppresses both action and endpoint steering in every state. Cosmos Policy supplies an earlier-model comparison in which effects are concentrated at early states and are most consistent with camera-visible robot motion. Together, the results support future-dependent action generation without establishing candidate search, outcome comparison, or task-level planning.

2 Related Work

Prediction and control. Some visual robot policies generate future frames and infer actions from them; others condition action inference on predicted video representations [1, 7]. Joint video-action models instead learn future observations and controls in one model [6, 8, 10, 14, 20]. Their training objective does not determine whether a particular generated future is used at inference. UVA, for example, can omit video generation at test time [10]. Explicit visual planners answer a different question by sampling alternatives, predicting their outcomes, and comparing them with a goal-dependent score [2, 3].

Recent WAM studies separate predictive training from the use of predictions during action generation. Fast-WAM removes explicit test-time rollout while retaining competitive performance in its evaluations [21]; Faster-WAM retains a lower-cost future-conditioning route and reports improved robustness [24]. AGRA finds that visually plausible foresight need not support reliable action inference in its studied architecture [16]. RIFT directly establishes dependence on future-position values and their organization [22]. Our contribution is complementary: we exchange two coherent, policy-generated alternatives and ask whether behavior follows the selected donor.

Mechanistic interventions. Activation replacement tests whether an internal representation affects an output rather than merely being decodable from it [12, 18]. Path patching extends this logic to specified computational routes [5]. Conclusions can depend on the counterfactual pair and metric [23], while multi-component effects need not add when patched and unpatched routes interact [17]. We therefore treat K/V replacement as evidence for an active pathway, not a complete circuit decomposition.

73 3 Methods

74 3.1 Future transplantation and directional projection

75 Let saved simulator state S admit two unmodified branches A and B , with future representations
 76 F_A, F_B , action chunks $\mathbf{a}_A, \mathbf{a}_B$, and endpoints $\mathbf{e}_A, \mathbf{e}_B$ obtained by executing one action chunk. We
 77 call A the recipient and B the donor. To recompute A , we restore the same state and observation,
 78 retain its instruction, action noise, future-noise path, solver, and denoising schedule, and replace
 79 only F_A with F_B . We also reverse the branch roles.

80 For an action or endpoint $\mathbf{x} \in \{\mathbf{a}, \mathbf{e}\}$, normalized donor projection is

$$\pi_x(\hat{\mathbf{x}}; A, B) = \frac{(\hat{\mathbf{x}} - \mathbf{x}_A)^\top (\mathbf{x}_B - \mathbf{x}_A)}{\|\mathbf{x}_B - \mathbf{x}_A\|_2^2}. \quad (1)$$

81 The recipient is 0 and the donor is 1; negative values move away from the donor and values above 1
 82 overshoot it. The primary steering contrast subtracts the corresponding self-future recomputation:

$$\Delta_x = \pi_x(\hat{\mathbf{x}}_{\text{donor}}; A, B) - \pi_x(\hat{\mathbf{x}}_{\text{self}}; A, B). \quad (2)$$

83 Pairs are selected using native action and endpoint separation, never intervention outcomes. Every
 84 endpoint arm is executed after restoring the exact simulator state.

85 3.2 Models, populations, and controls

86 **Cosmos Policy.** We evaluate the public `Cosmos-Policy-LIBERO-Predict2-2B` check-
 87 point in direct-policy mode on LIBERO-Long, also called LIBERO-10 [8, 11]. The frozen grid
 88 contains 30 states: ten tasks crossed with the first query, a state after three open-loop chunks, and
 89 a state after six chunks. Eight native runs determine one reciprocal donor pair per state; three
 90 future-noise draws are repeated within direction. Endpoints concatenate goal-relevant simulator
 91 coordinates with the official nine-dimensional proprioceptive endpoint.

92 **Cosmos 3.** We evaluate the 16B `Cosmos3-Nano-Policy-DROID` checkpoint on six eligible
 93 RoboLab tasks [14, 15, 19]. Its diffusion generator has 36 two-way full-attention layers, four
 94 denoising calls, 33 raw video frames mapped to nine VAE latent frames, and 32 absolute joint-
 95 position actions. The transplant targets unconditioned future-video coordinates at the final guided-
 96 sampler boundary. A frozen screen selected 24 state clusters, capped at four per task, using only
 97 native success, action and endpoint separation, and replay validity. Twenty-two completed. Two
 98 source branches ended before producing full 33-frame videos and were left missing rather than
 99 replaced. Reciprocal directions and predicted versus executed future sources are repeated measure-
 100 ments within state.

101 Controls include exact self-future recomputation, Gaussian targets matched to the donor in norm
 102 and recipient distance, a prespecified natural-future control, within-task shuffled futures, and bitwise
 103 no-op checks. For Cosmos 3, identity layout capture, sampler wrapping, cache record/replay, and
 104 self patches reproduce actions bitwise; sentinels confirm that the transplant does not directly write
 105 to action coordinates. Executed-future conditions encode video from reachable donor actions and
 106 replay the same environment prefix before intervention.

107 3.3 Robot and object interventions

108 We isolate content in complementary ways. In Cosmos Policy, first-query interventions replace the
 109 wrist-camera future, primary-camera future, or future proprioception separately. A second study
 110 crosses binary robot and object states in re-rendered videos at 20 held-out states. A decoder checks
 111 whether the intended cell remains visible. A natural-pair screen separately searches policy-generated
 112 continuations for robot-divergent/object-matched and object-divergent/robot-matched futures; only
 113 four states meet its criteria, below the planned minimum of ten.

114 For Cosmos 3, a separately frozen factor study selected 12 population states round-robin by native
 115 object-position separation. Ten completed across all six tasks. Starting from the recipient video, an
 116 object-priority disjoint mask copies donor object pixels, donor robot pixels outside the object mask,
 117 both, or neither. This 2×2 design tests isolated pixel factors; it does not guarantee an in-distribution
 118 video or isolate a latent semantic variable.

119 3.4 K/V interventions

120 In Cosmos Policy, action queries can attend to future keys. We continuously gate future keys at
121 prespecified block 27, with exact all-key recomputation, equal-count random-key and current-key
122 controls, and a block-0 control. This removal changes attention normalization, so it tests whether a
123 late future-to-action edge is active, not the information carried by a token-count-preserving route.

124 Cosmos 3 permits the stronger replacement test. We record future-video keys and values from a
125 self-future run, then substitute them for donor-run future-token K/V at the frozen direct interface.
126 Token count, token order, current-video K/V, text K/V, action K/V, observation, instruction, and
127 noise remain fixed. Layer and pathway choices were frozen on excluded calibration tasks before
128 the 22-state population intervention. We report *K/V replacement loss*: the reduction in donor projec-
129 tion after substituting self-future K/V. This is an operational intervention contrast, not a population
130 natural indirect effect.

131 3.5 Statistical analysis

132 The independent unit is a restored simulator state. We first average reciprocal directions, donors,
133 noise draws, and process repeats within state, then form 95% percentile intervals from 10,000
134 bootstrap resamples of states [4]. Task-balanced sensitivity analyses average states within task be-
135 fore resampling tasks. We prespecified 0.10 projection units—10% of the native recipient-to-donor
136 displacement—as the smallest effect of interest for the Cosmos Policy confirmatory family and the
137 Cosmos 3 population and factor families [9]. Prespecified primary sign-test p -values use Holm
138 correction.

139 4 Results

140 4.1 Do coherent futures steer behavior?

141 **Yes: coherent donor futures steer both actions and their one-chunk simulator endpoints.**

142 **Cosmos 3.** Across 22 completed states and all six tasks, predicted-future action steering is 0.992
143 (95% state-bootstrap CI [0.972, 1.010]), positive in 22/22 states. It is 0.433 ([0.332, 0.538]) above
144 the natural-future control. Futures encoded from executed donor branches yield action steering
145 of 0.983 ([0.890, 1.064]) and executed-endpoint steering of 1.003 ([0.950, 1.050]), both positive in
146 22/22 states. Executed-future action steering exceeds the geometry-matched Gaussian by 0.941
147 ([0.837, 1.032]) and the natural control by 0.391 ([0.246, 0.533]). Task-balanced analyses give the
148 same qualitative conclusions. A separate four-task engineering study identifies the correct donor in
149 12/12 cases for both future sources.

150 **Cosmos Policy.** At the first query, donor-minus-self action steering is 0.499 ([0.336, 0.679]), pos-
151 itive in all ten tasks. Executed-endpoint steering is 0.552 ([0.387, 0.728]), also positive in all ten.
152 The action contrast remains positive against Gaussian (0.497), natural (0.103), and shuffled-future
153 (0.255) controls; endpoint contrasts are 0.549, 0.110, and 0.277. Secondary control intervals were
154 not available. Steering is much smaller at later sampled states: after three chunks, action and end-
155 point effects are 0.031 and 0.004; after six, they are 0.027 and 0.025 (Figure 2). Because each time
156 point uses a different state, this is a state/timing association rather than a causal estimate of temporal
157 decay.

158 No screened branch pair in either policy retained a stable success/failure label across continuation
159 seeds. The result concerns donor-directed action and one-chunk endpoint changes, not closed-loop
160 task completion.

161 4.2 What future content causes steering?

162 **The content pattern differs across checkpoints: Cosmos Policy favors visual robot-motion in-**
163 **formation, while Cosmos 3’s whole-future effect is not recreated by either isolated pixel factor.**

164 In Cosmos Policy, wrist-future action steering is 0.435 ([0.284, 0.605]), primary-camera steering is
165 0.101 ([0.057, 0.144]), and future-proprioception steering is 0.001 ([−0.004, 0.008]). In the 20-state

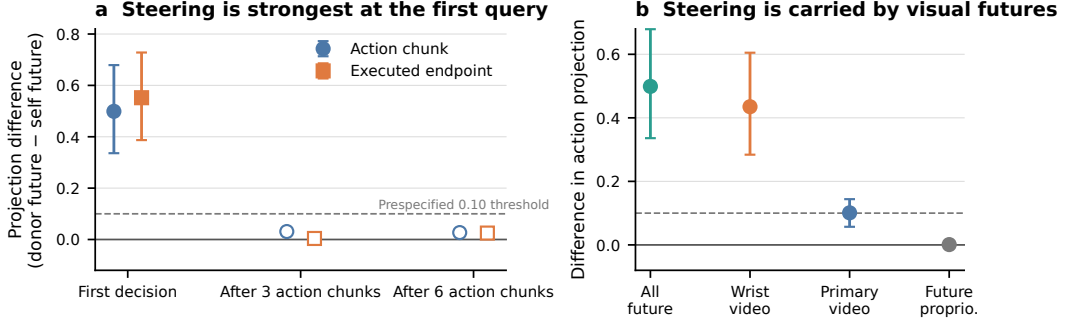


Figure 2: Cosmos Policy results. (a) Donor-minus-self action and endpoint steering is large at first-query states and small at later registered states. (b) At first-query states, visual futures carry the action effect, whereas future proprioception does not. Points are state means; bars are 95% state-bootstrap intervals where reported. The dashed line is the prespecified 0.10 threshold.

2 × 2 study, decoded futures identify the intended cell with 92.5% primary-camera and 98.75% wrist-camera accuracy, yet the object-state main effect is 0.00015 ([-0.00012, 0.00049]) for action and 0.00018 ([-0.00040, 0.00102]) for the executed goal-state endpoint. The available analysis does not report the robot-state main effect. In the four exploratory natural pairs, robot-motion futures yield action steering of 0.291 ([0.060, 0.521]) and robot-endpoint steering of 0.315 ([0.059, 0.571]); object-only effects are 0.021 for both action and goal endpoint, with intervals crossing zero. Together, these interventions are consistent with an inverse-dynamics-like role for camera-visible robot motion at the sampled early states, but do not show that Cosmos Policy generally ignores object consequences.

Cosmos 3 provides a counterpoint. Across ten factor-study states and all six tasks, the isolated robot-pixel effect is 0.0022 ([-0.0016, 0.0060]) for action and 0.0102 ([-0.0067, 0.0275]) for the full executed endpoint. The isolated object-pixel effect is 0.0061 ([0.0009, 0.0119]) for action and -0.0001 ([-0.0254, 0.0239]) for the endpoint. Every interval lies inside the prespecified [-0.10, 0.10] region, and task-balanced estimates agree. Thus neither isolated factor recreates a meaningful fraction of coherent steering (Figure 3c). This is consistent with information distributed across the coherent future or with nonlinear interactions among content, masks, and tokenization; it does not show that the two factors are jointly necessary.

4.3 Which internal pathway carries the effect?

Future-token attention provides an active route from generated futures to action, with the strongest evidence coming from population-level K/V replacement in Cosmos 3.

In Cosmos Policy, block-27 future-key removal produces a monotonic action-disruption curve: normalized L_2 change is 0.022, 0.035, 0.054, and 0.068 at gates 0.25, 0.50, 0.75, and 1.00. All-key recomputation is bitwise exact. Full removal is 0.016 more disruptive than equal-count random-key removal and 0.048 more disruptive than block-0 future-key removal; equal-count current-key removal is 0.024 larger. Executed endpoints support the same bounded conclusion, although state counts and intervals are not available for this analysis. The late future-to-action edge is active, but it is neither the only nor the largest route through the block.

In Cosmos 3, the interface chosen on excluded calibration tasks was frozen before population testing. Replacing donor-run future-token K/V with self-future K/V reduces predicted-future action projection by 0.877 ([0.842, 0.910]), positive in 22/22 states. For executed futures, it reduces action projection by 0.814 ([0.723, 0.900]) and executed-endpoint projection by 0.851 ([0.766, 0.922]), again positive in every state. Task-balanced point estimates are 0.877, 0.812, and 0.853; all prespecified primary sign tests remain significant after Holm correction. Because the replacement keeps token count fixed and does not mask attention positions, it provides stronger evidence that the future-token K/V interface carries donor-directed information. It does not establish a unique or complete circuit.

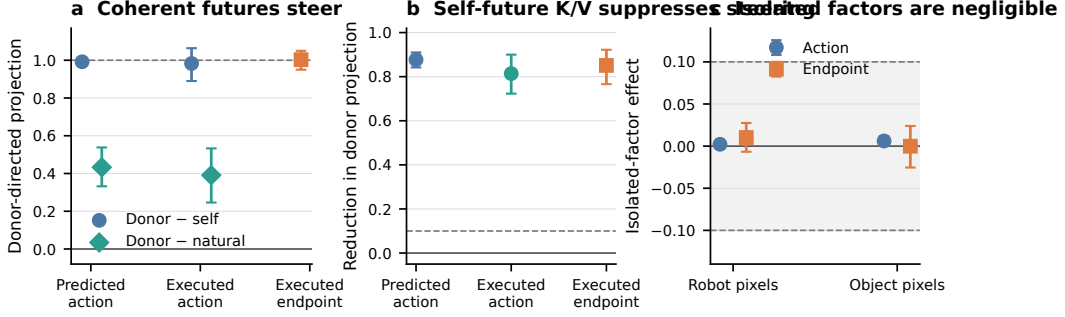


Figure 3: Cosmos 3 population results. (a) Coherent predicted and executed futures steer actions, outperform natural controls, and move one-chunk simulator endpoints toward the donor across 22 states and six tasks. (b) Replacing donor-run future-token K/V with self-future K/V removes most donor-directed action and endpoint projection. (c) In ten factor-study states, isolated robot- and object-pixel effects lie inside the prespecified $[-0.10, 0.10]$ region. Bars are 95% state-bootstrap intervals.

5 Discussion and Limitations

Together, the experiments support future-dependent action generation across two WAM generations, but not a single shared content code. A coherent donor future directs the action toward its associated alternative, and the endpoint follows after simulator execution. This is more specific than showing that generic corruption changes behavior: the intervention predicts which alternative the model will approach.

The content results narrow the interpretation. Cosmos Policy’s early-state pattern resembles an inverse-dynamics interface conditioned on visible robot motion. Cosmos 3 does not admit the same isolated-pixel account: strong whole-future steering coexists with negligible robot-only and object-only effects. The difference is inconsistent with a model-independent robot-pixel explanation and suggests that usable content can depend on architecture, context, or distributed feature interactions. The pathway experiments complement this finding. Cosmos Policy identifies an active late attention edge; Cosmos 3 shows across 22 states that future-token K/V carries most of the donor-directed shift.

The evidence remains narrower than planning. We do not test candidate generation, outcome comparison, or value-based selection, and we do not establish effects on binary task success. “Future” and “imagined” describe generated representations, not deliberation. The claim is therefore that these direct policies use coherent futures to condition action, not that WAMs generally plan over task consequences.

Limitations. All interventions are in simulation and cover two checkpoints, six RoboLab tasks, and LIBERO-Long. Cosmos Policy’s strongest result uses one first-query state per task, while later estimates use different states and cannot isolate elapsed time. Its natural content analysis contains only four states, and rendered hybrids may be outside the policy’s native future distribution. Cosmos 3 completed 22 of 24 frozen population states and 10 of 12 factor states; the two population omissions resulted from short source videos and were not replaced. Pixel masks may miss distributed features or induce nonlinear changes, so negligible isolated effects do not establish joint necessity. Finally, key removal changes attention normalization, and K/V replacement can interact with unpatched routes. These interventions identify active pathways, not a full circuit.

Broader impact. Mechanistic tests can help distinguish plausible-looking predictions from representations that actually control behavior. The same interfaces could also redirect robot actions or create false confidence if validated on narrow state distributions. Use beyond simulation would require broader coverage, physical safety constraints, and monitoring outside the intervention metric.

6 Conclusion

Coherent generated futures steer actions and one-chunk simulator endpoints in two WAM checkpoints. In Cosmos 3, the result holds across 22 sampled states, is strongly attenuated by future-token K/V replacement, and is not recreated by isolated robot or object pixels. Cosmos Policy shows a related but more state-dependent pattern that favors visual robot-motion information. These results identify future-conditioned action generation while leaving general planning claims open.

References

- [1] Yilun Du, Sherry Yang, Bo Dai, Hanjun Dai, Ofir Nachum, Joshua B. Tenenbaum, Dale Schuurmans, and Pieter Abbeel. Learning universal policies via text-guided video generation. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/1d5b9233ad716a43be5c0d3023cb82d0-Abstract.html.
- [2] Yilun Du, Sherry Yang, Pete Florence, Fei Xia, Ayzaan Wahid, Brian Ichter, Pierre Sermanet, Tianhe Yu, Pieter Abbeel, Joshua B. Tenenbaum, Leslie Kaelbling, Andy Zeng, and Jonathan Tompson. Video language planning. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=9pKtcJcMP3>.
- [3] Frederik Ebert, Chelsea Finn, Sudeep Dasari, Annie Xie, Alex Lee, and Sergey Levine. Visual foresight: Model-based deep reinforcement learning for vision-based robotic control. *arXiv preprint arXiv:1812.00568*, 2018. URL <https://arxiv.org/abs/1812.00568>.
- [4] C. A. Field and A. H. Welsh. Bootstrapping clustered data. *Journal of the Royal Statistical Society: Series B*, 69(3):369–390, 2007. doi: 10.1111/j.1467-9868.2007.00593.x.
- [5] Nicholas Goldowsky-Dill, Chris MacLeod, Lucas Sato, and Aryaman Arora. Localizing model behavior with path patching, 2023. URL <https://arxiv.org/abs/2304.05969>. arXiv preprint, version 2.
- [6] Yanjiang Guo, Yucheng Hu, Jianke Zhang, Yen-Jen Wang, Xiaoyu Chen, Chaochao Lu, and Jianyu Chen. Prediction with action: Visual policy learning via joint denoising process. In *Advances in Neural Information Processing Systems*, volume 37, pages 112386–112410. Curran Associates, Inc., 2024. doi: 10.52202/079017-3570. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/cbe25fa0e7c7084049276888a09acc8d-Abstract-Conference.html.
- [7] Yucheng Hu, Yanjiang Guo, Pengchao Wang, Xiaoyu Chen, Yen-Jen Wang, Jianke Zhang, Koushil Sreenath, Chaochao Lu, and Jianyu Chen. Video prediction policy: A generalist robot policy with predictive visual representations. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 24328–24346. PMLR, 2025. URL <https://proceedings.mlr.press/v267/hu25g.html>.
- [8] Moo Jin Kim, Yihuai Gao, Tsung-Yi Lin, Yen-Chen Lin, Yunhao Ge, Grace Lam, Percy Liang, Shuran Song, Ming-Yu Liu, Chelsea Finn, and Jinwei Gu. Cosmos Policy: Fine-tuning video models for visuomotor control and planning. In *The Fourteenth International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=wPEIStHxYH>.
- [9] Daniël Lakens. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362, 2017. doi: 10.1177/1948550617697177.
- [10] Shuang Li, Yihuai Gao, Dorsa Sadigh, and Shuran Song. Unified video action model, 2025. URL <https://arxiv.org/abs/2503.00200>. arXiv preprint, version 3.
- [11] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. In *Advances in Neural Information Processing Systems*, volume 36, pages 44776–44791. Curran Associates, Inc., 2023. doi: 10.52202/075280-1939. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/8c3c666820ea055a77726d66fc7d447f-Abstract-Datasets_and_Benchmarks.html.
- [12] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372. Curran Associates, Inc., 2022. doi: 10.52202/068431-1262. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/6f1d43d5a82a37e89b0665b33bf3a182-Abstract-Conference.html.

- [13] Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. RoboCasa: Large-scale simulation of household tasks for generalist robots. In *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, July 2024. doi: 10.15607/RSS.2024.XX.050. URL <https://www.roboticsproceedings.org/rss20/p050.html>.
- [14] NVIDIA. Cosmos 3: Omnimodal world models for physical AI. *arXiv preprint arXiv:2606.02800*, 2026. doi: 10.48550/arXiv.2606.02800. URL <https://research.nvidia.com/labs/cosmos-lab/cosmos3/technical-report.pdf>.
- [15] NVIDIA. Cosmos3-nano-policy-droid. Hugging Face model checkpoint and configuration, 2026. URL <https://huggingface.co/nvidia/Cosmos3-Nano-Policy-DROID>. Checkpoint revision 2b9f9517.
- [16] Lu Qiu, Yizhuo Li, Yi Chen, Yuying Ge, Yixiao Ge, and Xihui Liu. Making foresight actionable: Repurposing representation alignment in world action models, 2026. URL <https://arxiv.org/abs/2606.12217>. arXiv preprint.
- [17] Sankaran Vaidyanathan, David Arbour, Aaron Mueller, Scott Niekum, and David Jensen. The curse of multiple mediators: Hidden interaction effects in activation patching, 2026. URL <https://arxiv.org/abs/2606.27510>. arXiv preprint.
- [18] Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, volume 33, pages 12388–12401. Curran Associates, Inc., 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/92650b2e92217715fe312e6fa7b90d82-Abstract.html>.
- [19] Xuning Yang, Rishit Dagli, Alex Zook, Hugo Hadfield, Ankit Goyal, Stan Birchfield, Fabio Ramos, and Jonathan Tremblay. RoboLab: A high-fidelity simulation benchmark for analysis of task generalist policies. In *Proceedings of Robotics: Science and Systems*, Sydney, Australia, July 2026. doi: 10.15607/RSS.2026.XXII.096. URL <https://roboticsproceedings.org/rss22/p096.html>.
- [20] Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shen Yuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, You Liang Tan, Chuning Zhu, Jiannan Xiang, Ayaan Malik, Kyungmin Lee, William Liang, Nadun Ranawaka, Jiasheng Gu, Yinzhen Xu, Guanzhi Wang, Fengyuan Hu, Avnish Narayan, Johan Bjorck, Jing Wang, Gwanghyun Kim, Dantong Niu, Ruijie Zheng, Yuqi Xie, Jimmy Wu, Qi Wang, Ryan Julian, Danfei Xu, Yilun Du, Yevgen Chebotar, Scott Reed, Jan Kautz, Yuke Zhu, Linxi “Jim” Fan, and Joel Jang. World action models are zero-shot policies, 2026. URL <https://arxiv.org/abs/2602.15922>. arXiv preprint.
- [21] Tianyuan Yuan, Zibin Dong, Yicheng Liu, and Hang Zhao. Fast-WAM: Do world action models need test-time future imagination?, 2026. URL <https://arxiv.org/abs/2603.16666>. arXiv preprint, version 2.
- [22] Chushan Zhang, Jinguang Tong, Xuesong Li, Yikai Wang, and Hongdong Li. Keep the future, drop the rollout: RIFT for world action models, 2026. URL <https://arxiv.org/abs/2608.11521>. arXiv preprint, version 2.
- [23] Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=Hf17y6u9BC>.
- [24] Weiheng Zhao, Haoyi Jiang, Xin Shi, Liu Liu, Fan Huang, Zhizhong Su, Wei Sui, and Xinggang Wang. Faster-WAM: Efficient inference-time future conditioning for robust world action models, 2026. URL <https://arxiv.org/abs/2608.04404>. arXiv preprint.

A Reproducibility Details

The Cosmos Policy experiments use public commit 18a2acca, checkpoint revision cb689ec0, LIBERO, PyTorch 2.7 with CUDA 12.8, and two NVIDIA H100 80GB GPUs. The frozen design contains ten tasks, three state indices, eight native policy runs per state, one reciprocal donor pair per state, and three future-noise draws per direction. Confidence intervals use 10,000 bootstrap resamples of simulator states.

The Cosmos 3 experiments pin Cosmos commit 9aa98e5a, Cos-
mos Framework commit d4599e2e, RoboLab commit 9db0aaf0, and

337 checkpoint revision 2b9f9517. The frozen manifest has SHA-256
338 46b6fe3d6abb07712d0926361587b853b2e8313a10c7faeeccb0150d030e986b.
339 Endpoint evaluation reconstructs the environment, restores the saved state, and replays the ob-
340 servation prefix. Separate controls test token layout, sampler wrapping, cache record and replay,
341 self replacement, and the action-coordinate boundary. Complete timing and aggregate compute
342 accounting remain to be added before submission.

343 **B Additional Results and Boundaries**

344 Across all 30 Cosmos Policy states, mean donor-minus-self steering is 0.186 ([0.095, 0.290]) for
345 actions and 0.194 ([0.091, 0.305]) for endpoints. Because the three state indices may be nested
346 within task trajectories, the first-query analysis is the cleaner confirmatory result.

347 The six-state RoboCasa replication is exploratory and is not pooled with LIBERO [13]. Its donor-
348 minus-self endpoint effect is 0.0287 ([0.0101, 0.0492]), positive in 6/6 states; action steering is
349 smaller and does not reliably exceed the Gaussian control. In the Cosmos 3 population, the two
350 missing frozen states ended after 21 and 29 actions, before producing the 33-frame source video.
351 Neither was replaced. The prespecified minimum of 20 states across five tasks and the factor-study
352 minimum of ten states across five tasks were both met across six tasks.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract, Introduction, Results, and Discussion state the future-transplantation result, its variation across decision points, and the limits on claims about planning and task success.

Guidelines:

- The answer [\[N/A\]](#) means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [\[No\]](#) or [\[N/A\]](#) answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Result-specific qualifications appear beside the relevant findings, and the Discussion and Limitations section addresses model and state coverage, the later-state comparison, the factorized object intervention, the internal replacement experiment, and the absence of task-success results.

Guidelines:

- The answer [\[N/A\]](#) means that the paper has no limitation while the answer [\[No\]](#) means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper reports empirical intervention results and does not claim a theorem or formal proof.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [No]

Justification: The Methods, Results, and Appendix A specify the estimands, experimental units, controls, clustering, public checkpoints, and exactness gates. The complete seed list, all repository hashes and container digests, intervention code, and run logs are not yet included.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

- (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The source package includes machine-readable values for the paper figures, but not the intervention code, raw logs, or an anonymous public reproduction artifact.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: The Methods and Results provide the state grid, branch construction, intervention targets, controls, action horizons, model interfaces, and statistical analysis needed to interpret the results. Appendix A states which exact environment details are available and which remain unreported.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Principal estimates have explicitly defined 95% cluster-bootstrap intervals, the independent unit and 10,000-resample procedure are stated in the Methods, and exact sign counts are reported where relevant. The text identifies controls and secondary means for which exact intervals were not available.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [No]

Justification: Appendix A reports GPU type and software environment, but complete per-run wall-clock and aggregate compute-hour accounting is still being assembled for the submission artifact.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The experiments use public robot-policy checkpoints and simulation benchmarks, involve no people or personal data, and report negative and failed intervention results rather than selectively discarding them.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.

- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The Discussion and Limitations section discusses auditability benefits, action-redirecting misuse, false confidence from narrow simulation, and corresponding validation and safety-monitoring needs.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: This work does not release a new high-risk pretrained model or scraped dataset; it analyzes existing robot policies in simulation.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No]

Justification: Existing assets are cited and exact source revisions are recorded, but an explicit license inventory for every upstream code and benchmark asset remains to be added before submission.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [No]

Justification: New intervention code and result manifests are documented in the working artifact, but the anonymized release package and its final license are not yet attached to this draft.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The research contains no crowdsourcing and no human participants.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.

666 • According to the NeurIPS Code of Ethics, workers involved in data collection, cura-
667 tion, or other labor should be paid at least the minimum wage in the country of the
668 data collector.

669 **15. Institutional review board (IRB) approvals or equivalent for research with human**
670 **subjects**

671 Question: Does the paper describe potential risks incurred by study participants, whether
672 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
673 approvals (or an equivalent approval/review based on the requirements of your country or
674 institution) were obtained?

675 Answer: [N/A]

676 Justification: The research contains no human-subject data collection or interaction.

677 Guidelines:

678 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
679 with human subjects.

680 • Depending on the country in which research is conducted, IRB approval (or equiva-
681 lent) may be required for any human subjects research. If you obtained IRB approval,
682 you should clearly state this in the paper.

683 • We recognize that the procedures for this may vary significantly between institutions
684 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
685 guidelines for their institution.

686 • For initial submissions, do not include any information that would break anonymity
687 (if applicable), such as the institution conducting the review.

688 **16. Declaration of LLM usage**

689 Question: Does the paper describe the usage of LLMs if it is an important, original, or
690 non-standard component of the core methods in this research? Note that if the LLM is used
691 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
692 scientific rigor, or originality of the research, declaration is not required.

693 Answer: [N/A]

694 Justification: Language models were used only for writing and formatting assistance, not
695 as a component of the experimental method or scientific evaluation.

696 Guidelines:

697 • The answer [N/A] means that the core method development in this research does not
698 involve LLMs as any important, original, or non-standard components.

699 • Please refer to our LLM policy in the NeurIPS handbook for what should or should
700 not be described.